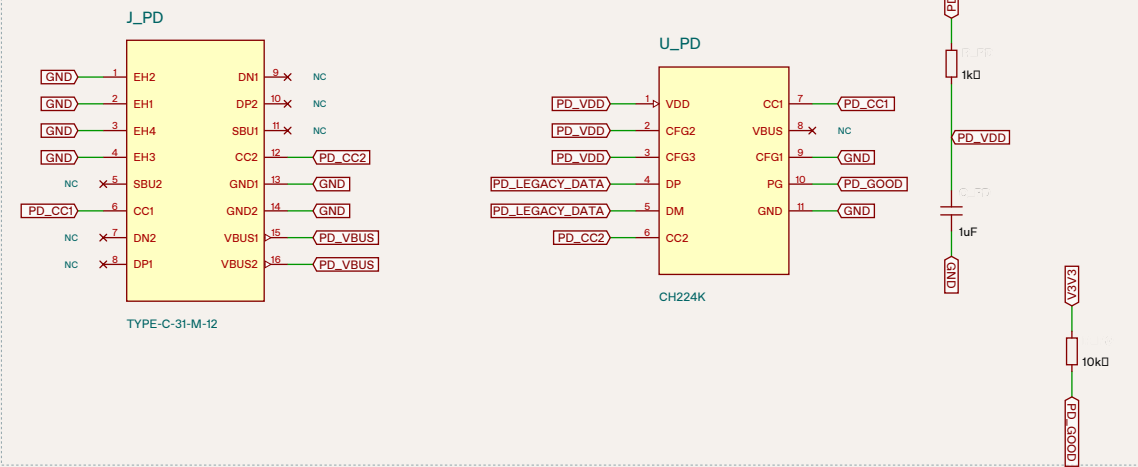


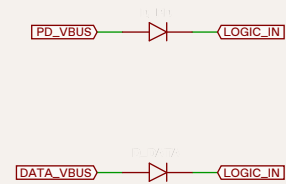
NEMA14 CH32X035G8U6 | USB-C PD & Logic Power

15 V USB-C PD negotiation



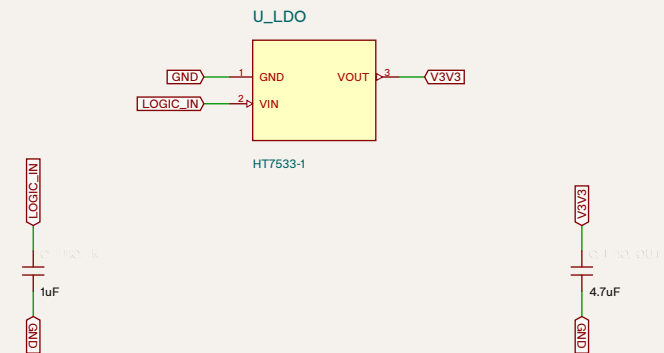
U_PD / CH224K requests 15 V (CFG1=0, CFG2=CFG3=1).
Open-drain PG is active low. DP/DM short: PD-only mode.
R_PD + C_PD supply/filter the negotiation IC.

USB / PD diode OR



Schottky diodes accept either supply and block crossfeed.

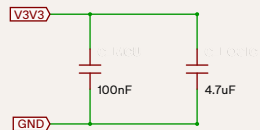
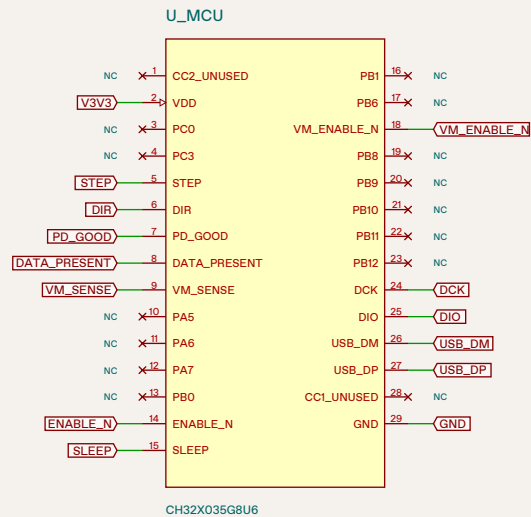
3.3 V logic regulator



U_LDO / HT7533-1 supplies the low-current CH32 logic.
30 V operating input, 33 V absolute maximum.
Input spikes and mounted thermal margins remain bench gates.

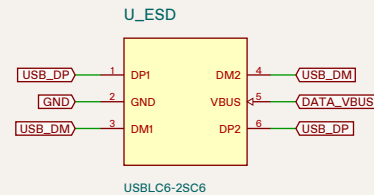
NEMA14 CH32X035G8U6 | CH32 & USB Control

Application MCU & local bypass

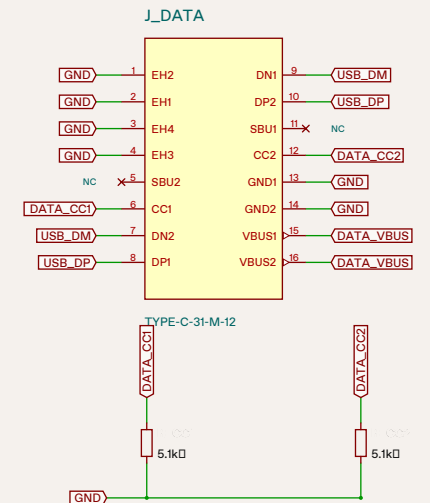


U_MCU / CH32X035 controls USB and STEP/DIR.
Internal clock avoids an external crystal.

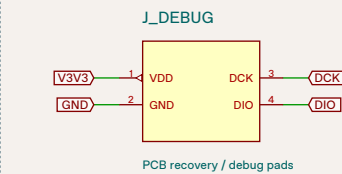
Computer USB-C & ESD protection



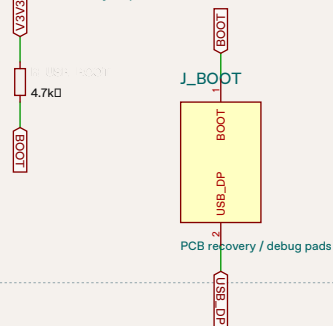
U_ESD / USBLC6 protects D+/D-, R_CCI/2 identify a USB sink.



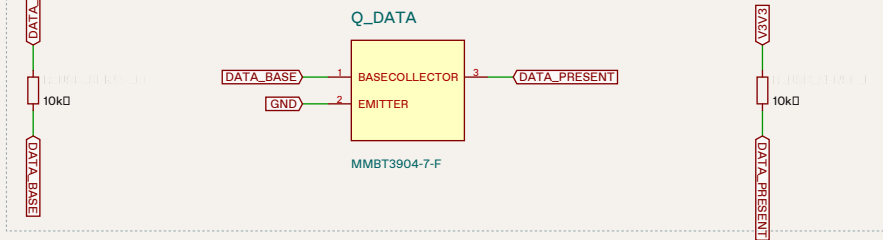
Programming & recovery



WCH debug pads and USB boot strap.
Use documented recovery sequence.



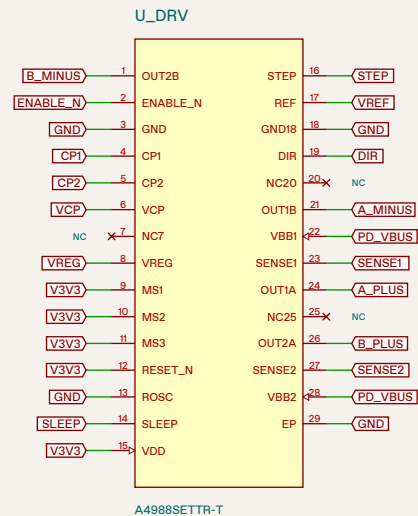
Host VBUS detection



Q_DATA / MMBT3904 senses host 5 V without a direct GPIO feed.
DATA_PRESENT LOW means host attached.

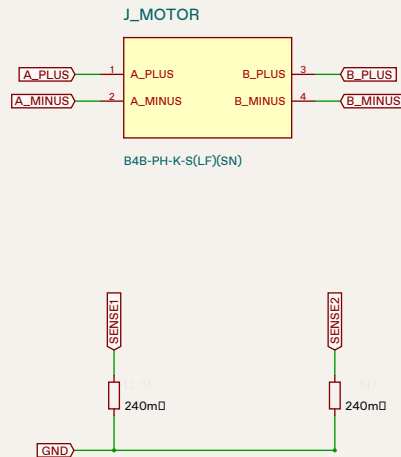
NEMA14 CH32X035G8U6 | Motor Driver & Voltage Monitor

A4988 / bipolar winding driver



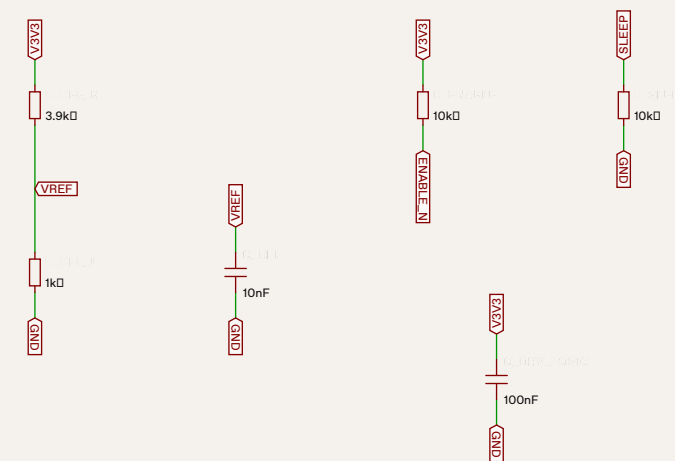
A4988SETTR-T

Keyed motor output & current shunts



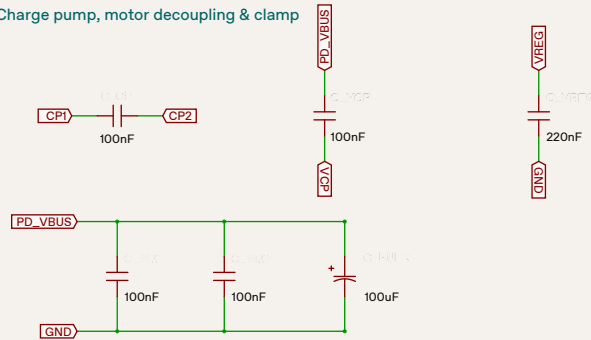
1 Black A+ 2 Green A-
3 Red B+ 4 Blue B-

Current reference & safe defaults



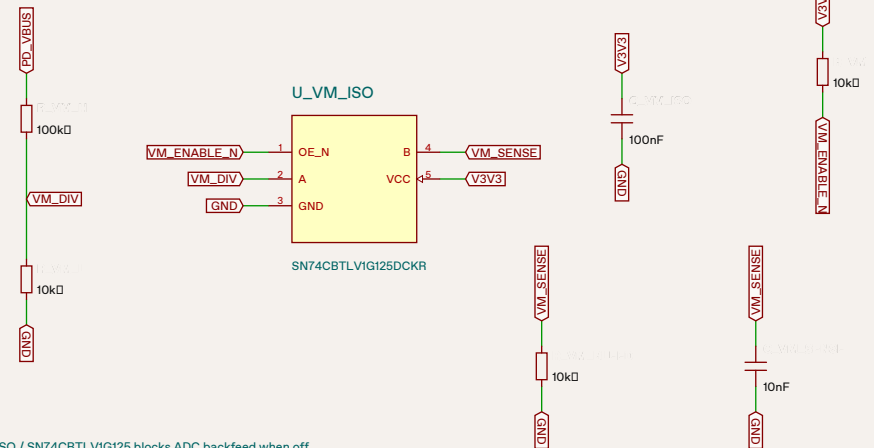
3.9k / 1k VREF and 0.24 ohm shunts: 0.351 A nominal.
ENABLE pull-up / SLEEP pull-down keep the driver off.

Charge pump, motor decoupling & clamp



U_DRV regulates both windings at fixed 1/16 microsteps.
C_CP / C_VCP / C_VREG support the internal driver rails.
Bulk capacitance and TVS limit transients; regeneration needs bench tests.

Motor-voltage monitor / power-off isolation



U_VM_ISO / SN74CBTLV1G125 blocks ADC backfeed when off.
OE pull-up defaults off; MCU enables, waits 0.5 ms, then samples VM/21.